# Bounded presentations of the symmetric groups on their standard generators with bit-length $O(\log n)$

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**Abstract**

We prove that there is an absolute constant $c$ such that, for every $n \geq 245$, the symmetric group $S_n$ has a presentation on its standard generators $(1,2)$ and $(1,2,\ldots,n)$ with at most 62 relators and length at most $c \log n$, where, as in Babai–Szemerédi and Bray–Conder–Leedham-Green–O'Brien, the length of a presentation is the number of symbols needed to write it down with exponents written in binary. This answers a question of Bray, Conder, Leedham-Green and O'Brien. The key observation is that if $p^2$ points are labelled row by row as the points of the affine plane $\mathbb{F}_p^2$, then the elementary transvections of $\mathrm{SL}(2,p)$ are represented by power words of length $O(\log n)$ in the standard generators; Carmichael's presentation of the alternating groups then allows us to use the affine group $\mathrm{ASL}(2,p)$ in place of the groups $\mathrm{PSL}(2,p)$ and $\mathrm{AGL}(1,p)$ used in earlier constructions.

## 1 Introduction

Following Babai and Szemerédi [1], the *length* of a presentation is the number of symbols needed to write it down, where each generator is one symbol and exponents are written in binary. This is the notion of length used by Bray, Conder, Leedham-Green and O'Brien [2]; Guralnick, Kantor, Kassabov and Lubotzky call it *bit-length* [6]. In [2, Thm. 1.2] it is shown that $S_n$ has a presentation with a bounded number of generators and relations and length $O(\log n)$, which is best possible because writing down $n$ already needs $\Omega(\log n)$ bits. These presentations contain the *standard generators* $(1,2)$ and $(1,2,\ldots,n)$, but also further generators. On the standard generators alone, [2, Thm. 1.1] gives presentations with $O(\log n)$ relations and length $O(\log^2 n)$. The authors of [2] state that they are left with one outstanding question:

*Does $S_n$ have a bounded presentation of length $O(\log n)$ on its standard generators?*

Replacing the extra generators by words in the standard generators, [2] obtains bounded presentations on the standard generators of length $O(n^2)$, and states that the existence of shorter ones is open. Guralnick, Kantor, Kassabov and Lubotzky proved that $S_n$ and $A_n$ have presentations with 3 generators, 7 relations and bit-length $O(\log n)$ [6, Thm. C] (Huxford [8] later identified errors in the arguments for this result and showed how to correct them). It is also shown in [6, Cor. 3.43] that for any two generators $a, b$ of $S_n$ there is a presentation on $a, b$ with 9 relations; for the standard pair its bit-length is $O(n \log^2 n)$ [6, Remark 3.44], and the authors state that they do not know whether $a, b$ can be chosen so as to obtain bit-length $O(\log n)$ [6, §1]. Theorem 1.1 below answers the question of [2] affirmatively; in the setting of [6], it shows that for $S_n$ the standard pair can be chosen, at the cost of a larger (but still bounded) number of relations. Our argument does not use the presentations of [6] or [8].

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**Theorem 1.1.** *There are absolute constants $r_0$ and $c_0$ such that for every $n \geq 2$ the symmetric group $S_n$ has a presentation on the generators $(1,2)$ and $(1,2,\ldots,n)$ with at most $r_0$ relators and length at most $c_0 \log n$. For $n \geq 245$ one can take $r_0 = 62$.*

**Idea of the proof.** The constructions of [2] and [6] start from a permutation group of degree about $p$ (for a prime $p$): $\mathrm{PSL}(2,p)$ acting on the projective line in [2], and $\mathrm{AGL}(1,p)$ (or its subgroup of index 2) acting on $\mathbb{F}_p$ in [6]. Such a group contains "multiplicative" maps such as $z \mapsto -1/z$ or $z \mapsto \lambda z$, and in any labelling of the points by consecutive integers these maps seem to have no short expression in the standard generators. We use instead the affine group $\mathrm{ASL}(2,p)$ acting on $\mathbb{F}_p^2$, with the $p^2$ points labelled row by row. The group $\mathrm{SL}(2,p)$ is generated by two elementary transvections, and the transvection $(a,b) \mapsto (a, a+b)$ only rotates the $a$-th row cyclically by $a$ places. The basic tool is the identity

$$\prod_{t=0}^{K-1} B^{S^t} \;=\; (BS^{-1})^K S^K,$$

valid in every group. Used twice with $S$ a power of the long cycle, it produces a product of row rotations in which row $r$ is rotated by $\min(r+1,\, 2p-1-r)$ places: on the first $p$ rows this is the transvection (up to a translation), and on the next $p$ rows it is the same transvection in reflected coordinates. We therefore let $\mathrm{ASL}(2,p)$ act simultaneously on two copies of the plane. Carmichael's presentation of the alternating groups only involves pairs of points, and $\mathrm{ASL}(2,p)$ has just four orbits on unordered pairs of points of the two planes; this yields a presentation of $S_{2p^2+2}$ with 27 relators on six generators, each of which is a power word of length $O(\log n)$ in the standard generators (Theorem 6.3). Conversely, the standard generators of $S_{2p^2+2}$ are short power words in these six generators (Proposition 6.4). Finally two overlapping copies of $S_{2p^2+2}$ are glued to obtain $S_n$ (Theorem 7.3).

**Remark 1.2.** (a) Theorem 1.1 concerns the notion of length of [1, 2], in which a power $E^e$ of a subword costs about $\log_2 |e|$ symbols for the exponent. In [5] a stronger notion is used (the total length of the relators as words in the free group); Theorem 1.1 says nothing about that notion.

(b) The constants are very large; we have made no attempt to optimise them (see Section 8).

(c) The relators are power expressions of bounded nesting depth (at most 17 in our construction).

## 1.1 Conventions

For integers $a \leq b$ we write $[a,b] = \{a, a+1, \ldots, b\}$. Permutations act on the *right*: the image of $\omega$ under $g$ is $\omega^g$, and $gh$ means "first $g$, then $h$", so $\omega^{gh} = (\omega^g)^h$. We write $g^h = h^{-1}gh$ and $[g,h] = g^{-1}h^{-1}gh$. With these conventions $(\omega_1,\ldots,\omega_k)^h = (\omega_1^h,\ldots,\omega_k^h)$, and $\mathrm{supp}(g)$ denotes the set of points moved by $g$.

We let $S_n = \mathrm{Sym}([0,n-1])$ and

$$x = (0,1), \qquad y = (0,1,2,\ldots,n-1),$$

so that $i^y = i+1$ for $i < n-1$ and $(n-1)^y = 0$. Under the bijection $i \mapsto i+1$ these are the standard generators $(1,2)$ and $(1,2,\ldots,n)$ of the symmetric group on $[1,n]$; hence it suffices to prove Theorem 1.1 for $x$ and $y$.

Vectors of $\mathbb{F}_p^2$ are *row* vectors and matrices act on the right, $v \mapsto vM$. For $M \in \mathrm{GL}(2,p)$ and $w \in \mathbb{F}_p^2$ let $\alpha_{M,w}$ be the affine map $v \mapsto vM+w$, and compose affine maps like permutations (first the left factor), so that $\alpha_{M,w}\alpha_{M',w'} = \alpha_{MM',wM'+w'}$. Then $\mathrm{ASL}(2,p) = \{\alpha_{M,w} : \det M = 1\}$,

its subgroup of translations is $N_T = \{t_w := \alpha_{I,w}\}$, and $\mathrm{SL}(2,p)$ is identified with $\{\alpha_{M,0}\}$. A direct computation gives

$$\alpha_{M,w}^{-1}\, t_u\, \alpha_{M,w} = t_{uM} \qquad (u \in \mathbb{F}_p^2). \tag{1}$$

We write $e_1 = (1,0)$, $e_2 = (0,1)$, and for $a \in \mathbb{F}_p$ we let $\overline{a} \in [0, p-1]$ denote its representative.

## 2 Power expressions, length, and presentations

**Definition 2.1.** Let $\mathcal{A}$ be a set (an alphabet). *Power expressions* (PEs) over $\mathcal{A}$ are defined recursively: every letter $a \in \mathcal{A}$ is a PE; if $E_1, \dots, E_k$ ($k \geq 2$) are PEs, so is the juxtaposition $E_1 E_2 \cdots E_k$; if $E$ is a PE and $e \in \mathbb{Z} \setminus \{0, 1\}$, then $(E)^e$ is a PE. We write $E^e$ for $(E)^e$, $E^{-1}$ for $(E)^{-1}$, and for PEs $E, F$ we use the PEs $E^F := F^{-1}EF$ and $[E,F] := E^{-1}F^{-1}EF$. A PE has an obvious value in any group once the letters are given values. The *length* $\ell(E)$ is defined by $\ell(a) = 1$ for $a \in \mathcal{A}$, $\ell(E_1 \cdots E_k) = \sum_i \ell(E_i)$, and

$$\ell\big((E)^e\big) = \ell(E) + 2 + \beta(e),$$

where $\beta(e)$ is the number of binary digits of $|e|$, plus 1 if $e < 0$. Thus $\ell(E)$ is the number of symbols of $E$ written with parentheses and binary exponents. We also write $\mathrm{occ}(E)$ for the number of occurrences of letters in $E$, $\mathrm{pw}(E)$ for the number of power operations $(\cdot)^e$ in $E$, and $\mathrm{ex}(E)$ for the largest $|e|$ occurring in $E$ (with $\mathrm{ex}(E) = 1$ if there is none).

Any other convention in which writing $(E)^e$ costs $\ell(E) + O(1 + \log|e|)$ symbols changes lengths by at most a constant factor; in particular the conventions of [1, 2] differ from ours by at most a constant factor.

**Lemma 2.2.** *For every PE $E$ we have $\ell(E) \leq \mathrm{occ}(E) + \mathrm{pw}(E)\,(\log_2 \mathrm{ex}(E) + 4)$.*

*Proof.* Each letter contributes 1 and each power $(\cdot)^e$ contributes $2 + \beta(e) \leq \log_2|e| + 4$. □

**Definition 2.3.** A set $\mathcal{E}$ of PEs is *uniform* if occ and pw are bounded on $\mathcal{E}$. By Lemma 2.2, there is then a constant $c_{\mathcal{E}}$ such that $\ell(E) \leq c_{\mathcal{E}}(1 + \log_2 \mathrm{ex}(E))$ for all $E \in \mathcal{E}$.

For a PE $E$ over $\mathcal{A}$ and PEs $W_a$ ($a \in \mathcal{A}$) over another alphabet $\mathcal{B}$, let $E[W]$ be the PE over $\mathcal{B}$ obtained by replacing every occurrence of each letter $a$ by $W_a$ (when $W_a$ is a juxtaposition placed inside a juxtaposition we simply concatenate). Clearly the value of $E[W]$ is the value of $E$ when each $a$ is given the value of $W_a$, and

$$\begin{gathered}\mathrm{occ}(E[W]) \leq \mathrm{occ}(E) \max_a \mathrm{occ}(W_a), \qquad \mathrm{pw}(E[W]) \leq \mathrm{pw}(E) + \mathrm{occ}(E) \max_a \mathrm{pw}(W_a), \\ \mathrm{ex}(E[W]) \leq \max\big(\mathrm{ex}(E), \max_a \mathrm{ex}(W_a)\big).\end{gathered} \tag{2}$$

Consequently, if $E$ ranges over a uniform set and the $W_a$ range over a uniform set, then the $E[W]$ range over a uniform set.

**Definition 2.4.** A *presentation* $\langle \mathcal{A} \mid \mathcal{R} \rangle$ consists of a finite alphabet $\mathcal{A}$ and a finite set $\mathcal{R}$ of PEs over $\mathcal{A}$ (the relators); it defines the group $F(\mathcal{A})/\langle\langle \mathcal{R} \rangle\rangle$, where $F(\mathcal{A})$ is the free group on $\mathcal{A}$ and the relators are evaluated in $F(\mathcal{A})$. Its length is $|\mathcal{A}| + \sum_{E \in \mathcal{R}} \ell(E)$. If $G$ is a group and $\varphi\colon \mathcal{A} \to G$ a map, we say that $\langle \mathcal{A} \mid \mathcal{R} \rangle$ is a presentation of $G$ *via* $\varphi$ (or *on* the elements $\varphi(a)$) if $\varphi$ extends to an isomorphism $F(\mathcal{A})/\langle\langle \mathcal{R} \rangle\rangle \to G$.

For a finite group $G$ this holds as soon as (i) every relator has value 1 under $\varphi$, (ii) $\varphi(\mathcal{A})$ generates $G$, and (iii) the group defined by the presentation has order at most $|G|$: indeed (i) and (ii) give a surjective homomorphism onto $G$, which is then bijective by (iii).

**Lemma 2.5** (Elimination of generators)**.** *Let $\langle \mathcal{A} \sqcup \mathcal{B} \mid \mathcal{R} \rangle$ be a presentation of a group $G$ via $\varphi$. Suppose that for every $b \in \mathcal{B}$ we are given a PE $W_b$ over $\mathcal{A}$ whose value under $\varphi|_{\mathcal{A}}$ is $\varphi(b)$. Put $W_a = a$ for $a \in \mathcal{A}$. Then $\langle \mathcal{A} \mid \{E[W] : E \in \mathcal{R}\} \rangle$ is a presentation of $G$ via $\varphi|_{\mathcal{A}}$.*

*Proof.* This is a sequence of Tietze transformations; we give the argument. Let $F = F(\mathcal{A} \sqcup \mathcal{B})$, $F' = F(\mathcal{A})$, $\mathcal{N} = \langle\langle \mathcal{R} \rangle\rangle_F$ and $\mathcal{N}' = \langle\langle E[W] : E \in \mathcal{R} \rangle\rangle_{F'}$. Let $\psi\colon F \to F'$ be the homomorphism with $\psi(a) = a$, $\psi(b) = W_b$, and $\iota\colon F' \to F$ the inclusion. Since $F/\mathcal{N} \cong G$ via $\varphi$ and $b^{-1}W_b$ has value 1 in $G$, we have $b^{-1}W_b \in \mathcal{N}$ for all $b \in \mathcal{B}$; hence $\iota\psi(f) \equiv f \pmod{\mathcal{N}}$ for all $f \in F$. In particular $\iota(E[W]) = \iota\psi(E) \in \mathcal{N}$ for $E \in \mathcal{R}$, so $\iota$ induces a homomorphism $\bar{\iota}\colon F'/\mathcal{N}' \to F/\mathcal{N}$; and $\psi(\mathcal{R}) \subseteq \mathcal{N}'$, so $\psi$ induces $\bar{\psi}\colon F/\mathcal{N} \to F'/\mathcal{N}'$. We have $\bar{\psi}\bar{\iota} = \mathrm{id}$ because $\psi\iota = \mathrm{id}_{F'}$, and $\bar{\iota}\bar{\psi} = \mathrm{id}$ because $\iota\psi(f) \equiv f \pmod{\mathcal{N}}$. Thus $F'/\mathcal{N}' \cong F/\mathcal{N} \cong G$, and the composite isomorphism sends $a \in \mathcal{A}$ to $\varphi(a)$. □

# 3 Short power words in $S_n$

**Lemma 3.1** (Telescoping)**.** *For elements $B, S$ of a group and an integer $K \geq 1$,*

$$\prod_{t=0}^{K-1} B^{S^t} = (BS^{-1})^K S^K,$$

*where the product on the left is taken in increasing order of $t$.*

*Proof.* For $K = 1$ both sides equal $B$. If the identity holds for $K$, then $(BS^{-1})^{K+1}S^{K+1} = (BS^{-1})^K S^K \cdot S^{-K}BS^{-1}S^{K+1} = \left(\prod_{t<K} B^{S^t}\right) B^{S^K}$. □

For PEs $B, S$ and $K \geq 1$ we write $\mathrm{Tel}(B, S, K)$ for the PE $(BS^{-1})^K S^K$ if $K \geq 2$, and $\mathrm{Tel}(B, S, 1) = B$. Note that $\mathrm{occ}(\mathrm{Tel}(B, S, K)) \leq \mathrm{occ}(B) + 2\mathrm{occ}(S)$ and $\mathrm{pw}(\mathrm{Tel}(B, S, K)) \leq \mathrm{pw}(B) + 2\mathrm{pw}(S) + 3$.

**Lemma 3.2.** *For elements $g, W$ of a group and $e \geq 1$ we have*

$$(gW)^e = g^e\, W^{g^{e-1}} W^{g^{e-2}} \cdots W^g W.$$

*Proof.* Induction on $e$:

$$(gW)^{e+1} = g^e W^{g^{e-1}} \cdots W \cdot gW = g^{e+1} \left(W^{g^{e-1}} \cdots W\right)^g W = g^{e+1} W^{g^e} \cdots W^g W. \qquad \square$$

**Lemma 3.3.** *For distinct points $z_0, \ldots, z_k$ we have*

$$(z_0, z_1)(z_1, z_2) \cdots (z_{k-1}, z_k) = (z_0, z_k, z_{k-1}, \ldots, z_1).$$

*Proof.* The point $z_0$ is moved successively to $z_1, z_2, \ldots, z_k$. For $1 \leq i \leq k$ the point $z_i$ is fixed by the first $i-1$ factors, mapped to $z_{i-1}$ by the $i$-th factor, and then fixed by the remaining factors, which only involve $z_i, \ldots, z_k$. □

**Lemma 3.4.** *Let $n \geq 3$ and let $x, y \in S_n$ be as in §1.1. Define PEs over $\{x, y\}$ by $P_1 = \sigma_1 = x$ and*

$$P_k = (xy^{-1})^k y^k \ (2 \leq k \leq n-1), \qquad \rho_L = P_{L-1}^{-1} \ (2 \leq L \leq n), \qquad \sigma_k = P_k P_{k-1}^{-1} \ (2 \leq k \leq n-1).$$

*(a)* $x^{y^i} = (i, i+1)$ *for* $0 \leq i \leq n-2$.

*(b)* $P_k = (0, k, k-1, \ldots, 1)$ *for* $1 \leq k \leq n-1$.

*(c)* $\rho_L = (0, 1, \ldots, L-1)$ *for* $2 \leq L \leq n$.

*(d)* $\sigma_k = (0,k)$ *for* $1 \le k \le n-1$.

*(e) Let* $0 \le s \le n-1$ *and* $g \in S_n$ *with* $\operatorname{supp}(g) \subseteq [0, n-1-s]$. *Then* $g^{y^s}$ *fixes* $[0, s-1]$ *and* $(i+s)^{g^{y^s}} = i^g + s$ *for* $i \in [0, n-1-s]$*; that is,* $g^{y^s}$ *is* $g$ *shifted by* $s$.

*Moreover the set of all these PEs, together with all* $y^s$ *and all PEs* $F^{y^s}$ *with* $F$ *in a fixed uniform set, is uniform.*

*Proof.* (a) $0^{y^i} = i$ and $1^{y^i} = i+1$.

(b) By Lemma 3.1 and (a), $P_k = \prod_{i=0}^{k-1} x^{y^i} = (0,1)(1,2)\cdots(k-1,k)$, which is $(0,k,k-1,\dots,1)$ by Lemma 3.3.

(c) This follows from (b).

(d) By (b),

$$P_k P_{k-1}^{-1} = (0,1)\cdots(k-2,k-1)\,(k-1,k)\,(k-2,k-1)\cdots(0,1) = (k-1,k)^h,$$

where $h = (k-2,k-1)\cdots(0,1)$; since $(k-1)^h = 0$ and $k^h = k$, this is $(0,k)$.

(e) For $i \in [0, n-1-s]$ we have $(i+s)^{y^{-s}} = i$, then $i^g \in [0, n-1-s]$, and $(i^g)^{y^s} = i^g + s$. For $j \in [0, s-1]$ we have $j^{y^{-s}} = j - s + n \in [n-s, n-1]$, which is fixed by $g$ because $\operatorname{supp}(g) \subseteq [0, n-1-s]$, and then mapped back to $j$ by $y^s$.

The last statement is clear, since the shapes of these PEs do not depend on $n, k, L, s$. □

# 4 Two planes and short transvections

**Setup 4.1.** Fix a prime $p \ge 5$ and put $q = p^2$ and $N = 2q+2$; let $n \ge N$. In $[0, n-1]$ we consider

$$G_0 = [0, q-1], \qquad G_1 = [q, 2q-1], \qquad \star = 2q, \qquad \infty = 2q+1,$$

and the bijections $\iota_0 \colon \mathbb{F}_p^2 \to G_0$ and $\iota_1 \colon \mathbb{F}_p^2 \to G_1$,

$$\iota_0(a,b) = \overline{a}\,p + \overline{b}, \qquad \iota_1(a,b) = q + \overline{a}\,p + \overline{-b}.$$

For $r \in [0, 2p-1]$ the *block* $R_r = [rp, rp+p-1]$ is row $r$ of $G_0$ if $r < p$, and $R_{p+u}$ $(0 \le u < p)$ is called row $u$ of $G_1$. (Thus $\iota_0(a,b)$ lies in row $\overline{a}$ of $G_0$ at position $\overline{b}$, and $\iota_1(a,b)$ lies in row $\overline{a}$ of $G_1$ at position $\overline{-b}$; the coordinates in $G_1$ are "reflected".) For $\alpha \in \mathrm{AGL}(2,p)$ let $\theta(\alpha) \in S_n$ be the permutation with

$$\iota_j(v)^{\theta(\alpha)} = \iota_j(v^\alpha) \quad (v \in \mathbb{F}_p^2,\ j = 0,1),$$

fixing all points of $[2q, n-1]$. Since we compose affine maps and permutations in the same order, $\theta$ is an injective homomorphism. Finally let

$$M_L = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad M_U = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \quad M_C = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad D_\lambda = \begin{pmatrix} \lambda & 0 \\ 0 & \lambda^{-1} \end{pmatrix} \ (\lambda \in \mathbb{F}_p^\times).$$

Thus $(a,b)^{\alpha_{M_L,0}} = (a, a+b)$, $(a,b)^{\alpha_{M_U,0}} = (a+b, b)$ and $(a,b)^{\alpha_{M_C,0}} = (b,-a)$. One checks directly that

$$M_L^{-1} M_U M_L^{-1} = M_C, \qquad M_C^{-1} M_L^{-1} M_C = M_U, \qquad M_L^\lambda M_U^{-\mu} M_L^\lambda M_C = D_\lambda \quad (\lambda\mu = 1). \tag{3}$$

(For the last identity: $M_L^\lambda M_U^{-\mu} M_L^\lambda = \begin{pmatrix} 0 & \lambda \\ -\mu & 0 \end{pmatrix}$.)

To describe permutations of $[0, 2q-1]$ we use two kinds of "rotations". For a function $m\colon [0,2p-1] \to \mathbb{Z}$ let $\mathrm{rot}(m)$ be the permutation that rotates each block $R_r$ by $m(r)$ places:

$$(rp+c)^{\mathrm{rot}(m)} = rp + \big((c+m(r)) \bmod p\big) \qquad (0 \le c < p),$$

fixing all points outside $[0, 2q-1]$. Next, for $c \in [0, 2p-1]$ and $i \in [0, p-1]$ let $d(c,i) = ip + c$ if $c < p$ and $d(c,i) = q + ip + (c-p)$ if $c \ge p$, and call $D_c = \{d(c,i) : 0 \le i < p\}$ the $c$-th *virtual column* (the columns of $G_0$, followed by the columns of $G_1$). For $m\colon [0,2p-1] \to \mathbb{Z}$ let $\mathrm{crot}(m)$ be the permutation with $d(c,i)^{\mathrm{crot}(m)} = d\big(c, (i+m(c)) \bmod p\big)$, fixing all points outside $[0, 2q-1]$. Clearly $\mathrm{rot}(m)\mathrm{rot}(m') = \mathrm{rot}(m+m')$ and $\mathrm{crot}(m)\mathrm{crot}(m') = \mathrm{crot}(m+m')$. We write $\mathbf{1}_X$ for the indicator function of a set $X$ and define the "tent" function

$$\tau(r) = \begin{cases} r+1, & 0 \le r \le p-1, \\ 2p-1-r, & p \le r \le 2p-1. \end{cases}$$

**Definition 4.2.** In the notation of Lemma 3.4, define the following PEs over $\{x, y\}$:

$$\begin{array}{llll} B = \mathrm{Tel}(\rho_p,\, y^p,\, p), & Z = \mathrm{Tel}(B,\, y^p,\, p), & B' = B^{y^q}, & L = Z\,B^{-1}B', \\ V = \mathrm{Tel}(\sigma_p,\, y^p,\, p-1)^{-1}, & B_c = \mathrm{Tel}(V,\, y,\, p), & \Phi = \rho_p\,\rho_p^{y^q}\,\sigma_q, & \Sigma = \mathrm{Tel}(\Phi,\, y^p,\, p), \\ Z_c = \mathrm{Tel}(B_c,\, \Sigma,\, p), & B'_c = B_c^{y^q}, & U = Z_c\,B_c^{-1}B'_c, & T = \rho_q^p\,(\rho_q^p)^{y^q}, \\ X_0 = \sigma_{2q}\,\sigma_{2q+1}, & X_1 = (\sigma_q\,\sigma_{q+1})^{y^q}, & Q = x^{y^{2q}}. & \end{array}$$

**Proposition 4.3.** *In Setup 4.1, the PEs of Definition 4.2 have the following values in $S_n$:*

*(a)* $B = \mathrm{rot}(\mathbf{1}_{[0,p-1]})$, $Z = \mathrm{rot}(\tau)$, $B' = \mathrm{rot}(\mathbf{1}_{[p,2p-1]})$;

*(b)* $V = (0, p, 2p, \dots, (p-1)p)$, $B_c = \mathrm{crot}(\mathbf{1}_{[0,p-1]})$, $\Phi = (0, 1, \dots, p-1, q, q+1, \dots, q+p-1)$, $d(c,i)^{\Sigma} = d\big((c+1) \bmod 2p,\, i\big)$ *for all* $c, i$, $Z_c = \mathrm{crot}(\tau)$, $B'_c = \mathrm{crot}(\mathbf{1}_{[p,2p-1]})$;

*(c)* $L = \theta(\alpha_{M_L,0})$, $U = \theta(\alpha_{M_U,0})$ *and* $T = \theta(t_{e_1})$;

*(d)* $X_0 = (0, \star, \infty)$, $X_1 = (q, \star, \infty)$ *and* $Q = (\star, \infty)$.

*Moreover, the set of all these PEs (for all $p$ and $n$) is uniform, and all their exponents are at most $2q+1 < n$ in absolute value. In particular each of them has length $O(\log n)$.*

*Proof.* Throughout we use Lemma 3.4; the hypothesis of part (e) of that lemma will be checked each time a conjugate by a power of $y$ is formed.

(a) $\rho_p = (0, \dots, p-1) = \mathrm{rot}(\mathbf{1}_{\{0\}})$. For $0 \le t \le p-1$ its support lies in $[0, n-1-pt]$, so $\rho_p^{y^{pt}} = \mathrm{rot}(\mathbf{1}_{\{t\}})$, and by Lemma 3.1 $B = \prod_{t<p} \mathrm{rot}(\mathbf{1}_{\{t\}}) = \mathrm{rot}(\mathbf{1}_{[0,p-1]})$. Since $\mathrm{supp}(B) = G_0 = [0, q-1]$ and $q - 1 + pt \le 2q - 1 \le n - 1$ for $0 \le t \le p$, we get $B^{y^{pt}} = \mathrm{rot}(\mathbf{1}_{[t,t+p-1]})$ for $0 \le t \le p$. In particular $B' = B^{y^q} = \mathrm{rot}(\mathbf{1}_{[p,2p-1]})$, and

$$Z = \prod_{t=0}^{p-1} \mathrm{rot}(\mathbf{1}_{[t,t+p-1]}) = \mathrm{rot}(m), \qquad m(r) = \#\{t \in [0, p-1] : t \le r \le t+p-1\}.$$

For $0 \le r \le p-1$ the admissible $t$ are $0, \dots, r$, and for $p \le r \le 2p-1$ they are $r-p+1, \dots, p-1$; hence $m = \tau$.

(b) For $0 \le t \le p-2$, Lemma 3.4 gives $\sigma_p^{y^{pt}} = (tp, (t+1)p)$, so by Lemmas 3.1 and 3.3, $\mathrm{Tel}(\sigma_p, y^p, p-1) = (0,p)(p,2p)\cdots((p-2)p, (p-1)p) = (0, (p-1)p, \dots, 2p, p)$, whence $V = (0, p, \dots, (p-1)p) = \mathrm{crot}(\mathbf{1}_{\{0\}})$. As $\mathrm{supp}(V) \subseteq [0, q-p]$, for $0 \le c \le p-1$ we get $V^{y^c} = \mathrm{crot}(\mathbf{1}_{\{c\}})$, and so $B_c = \mathrm{crot}(\mathbf{1}_{[0,p-1]})$ by Lemma 3.1.

Write $f_c = d(c,0)$, so $f_c = c$ for $c < p$ and $f_c = q + c - p$ for $c \geq p$. The permutation $\rho_p \rho_p^{y^q}$ maps $f_c \mapsto f_{c+1}$ for $c \notin \{p-1, 2p-1\}$, $f_{p-1} \mapsto f_0$ and $f_{2p-1} \mapsto f_p$, and $\sigma_q = (0,q) = (f_0, f_p)$; hence $\Phi$ maps $f_c \mapsto f_{(c+1) \bmod 2p}$, i.e. $\Phi$ is the stated $2p$-cycle. Since $\mathrm{supp}(\Phi) \subseteq [0, q+p-1]$ and $q+p-1+ip \leq 2q-1$ for $i \leq p-1$, the permutation $\Phi^{y^{ip}}$ is the cycle $(d(0,i), d(1,i), \ldots, d(2p-1,i))$, and these cycles for $0 \leq i \leq p-1$ are disjoint; by Lemma 3.1, $\Sigma$ is their product, which is the stated formula. Consequently, for every $m$,

$$\mathrm{crot}(m)^{\Sigma} = \mathrm{crot}\big(m(\,\cdot\, - 1)\big) \quad \text{(indices modulo } 2p),$$

because $\Sigma^{-1}\mathrm{crot}(m)\Sigma$ maps $d(c+1,i) = d(c,i)^{\Sigma}$ to $d(c, i+m(c))^{\Sigma} = d(c+1, i+m(c))$. Hence $B_c^{\Sigma^t} = \mathrm{crot}(\mathbf{1}_{[t,t+p-1]})$ for $0 \leq t \leq p-1$, and as in (a), $Z_c = \mathrm{crot}(\tau)$. Finally $\mathrm{supp}(B_c) = G_0$ and shifting by $q$ maps $d(c,i)$ to $d(c+p,i)$ for $c < p$, so $B'_c = \mathrm{crot}(\mathbf{1}_{[p,2p-1]})$.

(c) By (a), $L = \mathrm{rot}(\tau - \mathbf{1}_{[0,p-1]} + \mathbf{1}_{[p,2p-1]}) = \mathrm{rot}(\nu)$ with $\nu(r) = r$ for $r \leq p-1$ and $\nu(p+u) = p-u$ for $0 \leq u \leq p-1$. Now $\iota_0(a,b) = \overline{a}\,p + \overline{b} \in R_{\overline{a}}$ is moved by $\overline{a}$ places to $\overline{a}\,p + \overline{a+b} = \iota_0(a, a+b)$, and $\iota_1(a,b) = q + \overline{a}\,p + \overline{-b} \in R_{p+\overline{a}}$ is moved by $p - \overline{a} \equiv -a$ places to $q + \overline{a}\,p + \overline{-b-a} = \iota_1(a, a+b)$. Hence $L = \theta(\alpha_{M_L,0})$. Similarly, by (b), $U = \mathrm{crot}(\nu)$ with the same $\nu$. The point $\iota_0(a,b) = d(\overline{b}, \overline{a})$ is moved to $d(\overline{b}, \overline{a+b}) = \iota_0(a+b, b)$, and $\iota_1(a,b) = d(p + \overline{-b}, \overline{a})$ is moved by $p - \overline{-b} \equiv b$ to $d(p + \overline{-b}, \overline{a+b}) = \iota_1(a+b, b)$. Hence $U = \theta(\alpha_{M_U,0})$. Finally $\rho_q^p$ maps $i \mapsto (i+p) \bmod q$ on $[0, q-1]$, i.e. $\iota_0(a,b) \mapsto \iota_0(a+1, b)$; since $\mathrm{supp}(\rho_q^p) \subseteq [0, q-1] \subseteq [0, n-1-q]$, its conjugate by $y^q$ maps $q + \overline{a}\,p + v \mapsto q + \overline{a+1}\,p + v$, i.e. $\iota_1(a,b) \mapsto \iota_1(a+1, b)$. So $T = \theta(t_{e_1})$.

(d) For distinct $0, j, k$ we have $(0,j)(0,k) = (0,j,k)$. Hence $X_0 = (0, 2q, 2q+1)$, and $\sigma_q \sigma_{q+1} = (0, q, q+1)$, whose support lies in $[0, n-1-q]$ because $n \geq 2q+2$, so $X_1 = (q, 2q, 2q+1)$. Similarly $Q = (2q, 2q+1)$.

Uniformity: each PE in Definition 4.2 is obtained from $x$ and $y$ by a fixed sequence of operations (independent of $p$ and $n$), and all exponents occurring are among $\pm 1, p-1, p, q, 2q$ and the exponents occurring in $P_k$ for $k \leq 2q+1$. □

We shall also use the following PEs over the letters $L, U, T$, where $\lambda$ is a primitive root modulo $p$ and $\mu \in [1, p-1]$ satisfies $\lambda\mu \equiv 1 \pmod p$:

$$C := L^{-1} U L^{-1}, \qquad E := C\,T\,C^{-1}, \qquad D := L^{\lambda} U^{-\mu} L^{\lambda} C. \tag{4}$$

**Lemma 4.4.** *If $L, U, T$ are given the values $\alpha_{M_L,0}, \alpha_{M_U,0}, t_{e_1}$ in $\mathrm{ASL}(2,p)$ (or $\theta$ of these in $S_n$), then $C$, $E$ and $D$ have the values $\alpha_{M_C,0}$, $t_{e_2}$ and $\alpha_{D_\lambda,0}$ (respectively $\theta$ of these).*

*Proof.* For $C$ and $D$ this is (3). For $E$, by (1) with $\alpha = \alpha_{M_C,0}^{-1}$ we get $E = t_{e_1}^{\alpha} = t_{e_1 M_C^{-1}} = t_{e_2}$, since $e_1 M_C^{-1} = (1,0)\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = (0,1)$. □

## 5 Presentations of $\mathrm{SL}(2,p)$ and $\mathrm{ASL}(2,p)$

Throughout this section $p \geq 5$ is a prime, $h = (p+1)/2$, $\lambda$ is a primitive root modulo $p$ and $\mu \in [1, p-1]$ with $\lambda\mu \equiv 1 \pmod p$.

**Theorem 5.1** (Sunday [11]; see also [2, Prop. 3.3])**.** *For every odd prime $p$, the group*

$$\langle a, c \mid a^p,\ acacac^{-1},\ (a^h c a^4 c)^2 \rangle$$

*is isomorphic to $\mathrm{PSL}(2,p)$; in particular it has order $p(p^2-1)/2$.*

**Proposition 5.2.** *The presentation*

$$\langle a, c \mid a^p,\ acacac^{-1},\ (a^h c a^4 c)^2 c^{-2},\ c^4 \rangle$$

*is a presentation of $\mathrm{SL}(2,p)$ via $a \mapsto M_L$, $c \mapsto M_C$.*

*Proof. The relators hold.* $M_L^p = I$ and $M_C^2 = -I$, so $M_C^4 = I$. Next $M_L M_C = \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix}$ has trace 1 and determinant 1, so $(M_L M_C)^2 = M_L M_C - I$ and $(M_L M_C)^3 = (M_L M_C)^2 - M_L M_C = -I = M_C^2$; thus $M_L M_C M_L M_C M_L M_C^{-1} = (M_L M_C)^3 M_C^{-2} = I$. Finally $M_L^h M_C M_L^4 M_C = \begin{pmatrix} h & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 4 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 4h-1 & -h \\ 4 & -1 \end{pmatrix}$, which has determinant 1 and trace $4h - 2 \equiv 0$ (as $2h = p + 1 \equiv 1$); by the Cayley–Hamilton theorem its square is $-I = M_C^2$.

*Order.* Let $\hat{G}$ be the group presented. Put $z = c^2$. From $acacac^{-1} = 1$ we get $(ac)^3 = acacac = c^2 = z$, so $z$ commutes with $ac$ and with $c$, hence with $a$; thus $z$ is central, and $z^2 = c^4 = 1$. The quotient $\hat{G}/\langle z\rangle$ satisfies the three relators of Theorem 5.1, so $|\hat{G}/\langle z\rangle| \le p(p^2-1)/2$ and $|\hat{G}| \le p(p^2-1) = |\mathrm{SL}(2,p)|$. Since $M_L$ and $M_C$ generate $\mathrm{SL}(2,p)$ (the matrices $M_L^k$ and $M_C^{-1} M_L^{-k} M_C = M_U^k$, $k \in \mathbb{Z}$, are all the elementary transvections, and these generate $\mathrm{SL}(2,p)$), the claim follows from the remark after Definition 2.4. □

**Proposition 5.3.** *Let $\mathcal{G}$ be the group with generators $a, c, t$ and the nine relators*

$$\text{(A1) } a^p, \quad \text{(A2) } acacac^{-1}, \quad \text{(A3) } (a^h c a^4 c)^2 c^{-2}, \quad \text{(A4) } c^4, \quad \text{(A5) } t^p,$$
$$\text{(A6) } [t,u], \quad \text{(A7) } [t,\varepsilon], \quad \text{(A8) } t\,\varepsilon\,a^{-1}t^{-1}a, \quad \text{(A9) } d^{-1}t\,d\,t^{-\lambda},$$

*where $u := c^{-1}a^{-1}c$, $\varepsilon := ctc^{-1}$ and $d := a^\lambda u^{-\mu} a^\lambda c$. Then $\mathcal{G}$ is presented via $a \mapsto \alpha_{M_L,0}$, $c \mapsto \alpha_{M_C,0}$, $t \mapsto t_{e_1}$ as $\mathrm{ASL}(2,p)$.*

*Proof. The relators hold in* $\mathrm{ASL}(2,p)$. (A1)–(A4) hold by Proposition 5.2, since $M \mapsto \alpha_{M,0}$ is a homomorphism. By (3), $u \mapsto \alpha_{M_U,0}$ and $d \mapsto \alpha_{D_\lambda,0}$, and by (1), $\varepsilon \mapsto t_{e_1 M_C^{-1}} = t_{e_2}$. (A5) is clear. (A6): $t_{e_1}^{\alpha_{M_U,0}} = t_{e_1 M_U} = t_{e_1}$. (A7): translations commute. (A8): $t_{e_1} t_{e_2} = t_{e_1+e_2} = t_{e_1 M_L} = a^{-1}ta$. (A9): $d^{-1}td = t_{e_1 D_\lambda} = t_{\lambda e_1} = t^\lambda$. The images of $a, c, t$ generate $\mathrm{ASL}(2,p)$, because $M_L, M_C$ generate $\mathrm{SL}(2,p)$ and $N_T$ is the normal closure of $t_{e_1}$.

*Order.* Let $\mathcal{S} = \langle a, c\rangle \le \mathcal{G}$. By (A1)–(A4) and Proposition 5.2 there is a surjective homomorphism $\pi\colon \mathrm{SL}(2,p) \to \mathcal{S}$ with $\pi(M_L) = a$ and $\pi(M_C) = c$; then $\pi(M_U) = u$ and $\pi(D_\lambda) = d$ by (3). For $v \in \mathbb{F}_p^2 \setminus \{0\}$ choose $M \in \mathrm{SL}(2,p)$ with $e_1 M = v$ and put $t_v := t^{\pi(M)}$; also put $t_0 := 1$. This is well defined: if $e_1 M = e_1 M'$, then $M'M^{-1}$ fixes $e_1$, so $M'M^{-1} = M_U^k$ for some $k$, and $\pi(M_U^k) = u^k$ commutes with $t$ by (A6); hence $t^{\pi(M')} = t^{u^k \pi(M)} = t^{\pi(M)}$. We record:

(i) $t_{e_1} = t$, $t_{e_2} = t^{\pi(M_C^{-1})} = ctc^{-1} = \varepsilon$, and $t_{e_1+e_2} = t^{\pi(M_L)} = a^{-1}ta$;

(ii) $t_v^{\pi(M)} = t_{vM}$ for all $v$ and all $M \in \mathrm{SL}(2,p)$;

(iii) $t_v^p = 1$ for all $v$, by (A5).

*Scalars.* We have $t_{\lambda e_1} = t^{\pi(D_\lambda)} = d^{-1}td$, which equals the power $t^\lambda$ by (A9); and conjugating by $\pi(M)$ with $e_1 M = v$ gives $t_{\lambda v} = t_v^\lambda$ for all $v$. By induction $t_{\lambda^k v} = t_v^{\lambda^k}$; since $\lambda$ is a primitive root and by (iii), $t_{\delta v} = t_v^\delta$ for every $\delta \in \mathbb{F}_p^\times$ (where $t_v^\delta$ means $t_v^{\overline{\delta}}$).

*Commutation.* Let $v, w \in \mathbb{F}_p^2$. If $v, w$ are linearly dependent, then $t_v, t_w$ are powers of a common $t_{v'}$ by the previous paragraph, so they commute. Otherwise let $\Delta = \det\binom{v}{w} \neq 0$ and $w' = \Delta^{-1} w$. The matrix $M$ with rows $v$ and $w'$ lies in $\mathrm{SL}(2,p)$ and $e_1 M = v$, $e_2 M = w'$; by (i), (ii) and (A7), $[t_v, t_{w'}] = [t,\varepsilon]^{\pi(M)} = 1$. Since $t_w = t_{w'}^\Delta$, also $[t_v, t_w] = 1$.

*Additivity.* We show $t_{v+w} = t_v t_w$ for all $v, w$. This is clear if $v = 0$ or $w = 0$, and for $w = \delta v$ with $\delta \in \mathbb{F}_p^\times$ it follows from $t_{\delta v} = t_v^\delta$ and (iii). If $\det\binom{v}{w} = 1$, let $M$ have rows $v, w$; then (A8), (i) and (ii) give $t_v t_w = (t\,\varepsilon)^{\pi(M)} = (a^{-1}ta)^{\pi(M)} = t_{(e_1+e_2)M} = t_{v+w}$. In general, if $v, w$ are independent with $\det\binom{v}{w} = \Delta$, put $w' = \Delta^{-1} w$. For $1 \le k \le p-1$ we claim $t_{v+kw'} = t_v t_{w'}^k$: for $k = 1$ this is the case of determinant 1, and since $\det\binom{v+kw'}{w'} = 1$ and $v + kw' \neq 0$, we get $t_{v+(k+1)w'} = t_{v+kw'} t_{w'} = t_v t_{w'}^{k+1}$. Taking $k = \overline{\Delta}$ gives $t_{v+w} = t_v t_{w'}^\Delta = t_v t_w$.

Hence $v \mapsto t_v$ is a homomorphism from $(\mathbb{F}_p^2, +)$ to $\mathcal{G}$, and its image $\mathcal{N} = \{t_v\}$ has order at most $p^2$. By (ii) $\mathcal{N}$ is normalised by $\mathcal{S}$, and it is abelian and contains $t$; so $\mathcal{N} \trianglelefteq \mathcal{G} = \langle \mathcal{S}, t \rangle$ and $\mathcal{G} = \mathcal{N}\mathcal{S}$. Therefore $|\mathcal{G}| \le p^2 \, |\mathrm{SL}(2,p)| = |\mathrm{ASL}(2,p)|$, and the proposition follows from the remark after Definition 2.4. □

**Corollary 5.4.** *With $C, E, D$ as in (4), the following ten relators on the letters $L, U, T$ form a presentation of* $\mathrm{ASL}(2,p)$ *via $L \mapsto \alpha_{M_L,0}$, $U \mapsto \alpha_{M_U,0}$, $T \mapsto t_{e_1}$:*

$$L^p, \quad LCLCLC^{-1}, \quad (L^hCL^4C)^2C^{-2}, \quad C^4, \quad T^p, \quad [T,U], \quad [T,E],$$
$$TEL^{-1}T^{-1}L, \quad D^{-1}TDT^{-\lambda}, \quad U^{-1}LU^{-1}L^{-1}UL^{-1}.$$

*These relators form a uniform set, and their exponents are at most $p$ in absolute value.*

*Proof.* Start from the presentation of Proposition 5.3. Add a generator $U$ and the relator $U^{-1}c^{-1}a^{-1}c$; modulo this relator we may replace $u$ by $U$ in (A6) and (A9). The resulting presentation still presents $\mathrm{ASL}(2,p)$, now with $U \mapsto \alpha_{M_U,0}$. By (3) the PE $a^{-1}Ua^{-1}$ has the value $\alpha_{M_C,0}$ of $c$, so Lemma 2.5 allows us to eliminate $c$ by substituting $c \mapsto a^{-1}Ua^{-1}$. The relator $U^{-1}c^{-1}a^{-1}c$ becomes $U^{-1}(aU^{-1}a)a^{-1}(a^{-1}Ua^{-1})$, which is equal in the free group to $U^{-1}aU^{-1}a^{-1}Ua^{-1}$, and (A1)–(A9) become the first nine relators displayed. Renaming $a, t$ as $L, T$ gives the claim. Uniformity is clear. □

**Lemma 5.5.** *For $p \ge 5$ the normal subgroups of* $\mathrm{ASL}(2,p)$ *are* $1$, $N_T$, $N_T\langle \alpha_{-I,0} \rangle$ *and* $\mathrm{ASL}(2,p)$.

*Proof.* Let $\mathcal{M} \trianglelefteq \mathrm{ASL}(2,p)$. The subgroup $\mathcal{M} \cap N_T$ corresponds to an $\mathrm{SL}(2,p)$-invariant subgroup of $\mathbb{F}_p^2$; since $\mathrm{SL}(2,p)$ is transitive on $\mathbb{F}_p^2 \setminus \{0\}$, it is 0 or everything. If $\mathcal{M} \cap N_T = 1$, then $[\mathcal{M}, N_T] \subseteq \mathcal{M} \cap N_T = 1$, so $\mathcal{M}$ centralises $N_T$; by (1) an element $\alpha_{M,w}$ centralising $N_T$ has $uM = u$ for all $u$, i.e. $M = I$, so $\mathcal{M} \subseteq N_T$ and $\mathcal{M} = 1$. If $\mathcal{M} \supseteq N_T$, then $\mathcal{M}/N_T$ is a normal subgroup of $\mathrm{ASL}(2,p)/N_T \cong \mathrm{SL}(2,p)$. For $p \ge 5$ the group $\mathrm{PSL}(2,p)$ is simple and $\mathrm{SL}(2,p)$ is perfect (see e.g. [7, II.6]); so a normal subgroup of $\mathrm{SL}(2,p)$ not contained in $\{\pm I\}$ maps onto $\mathrm{PSL}(2,p)$, hence has index at most 2, hence equals $\mathrm{SL}(2,p)$. Thus $\mathcal{M}/N_T \in \{1, \{\pm I\}, \mathrm{SL}(2,p)\}$. □

## 6 A presentation of $S_{2p^2+2}$ on six short generators

**Theorem 6.1** (Carmichael [3]; see [4, §6.3])**.** *For $M \ge 1$, the group*

$$\langle x_1, \ldots, x_M \mid x_i^3 \ (1 \le i \le M), \ (x_ix_j)^2 \ (1 \le i < j \le M) \rangle$$

*is isomorphic to the alternating group $A_{M+2}$, via $x_i \mapsto (i, M+1, M+2)$.*

Note that $(x_jx_i)^2 = x_i^{-1}(x_ix_j)^2x_i$, so the relators $(x_ix_j)^2$ with $i \ne j$ in either order are consequences.

**Lemma 6.2.** *Let $\Omega$ be a finite set, $\star, \infty \notin \Omega$, and let $\omega_1, \ldots, \omega_r \in \Omega$ be distinct, $r \ge 2$.*

*(a) The permutation $\pi = (\omega_1, \star, \infty)(\omega_2, \star, \infty) \cdots (\omega_r, \star, \infty)$ maps $\omega_i \mapsto \omega_{i+2}$ for $i \le r-2$, $\omega_{r-1} \mapsto \infty$, $\omega_r \mapsto \star$, $\star \mapsto \omega_2$ and $\infty \mapsto \omega_1$, and fixes all other points.*

*(b) Suppose $r$ is even, $g \in \mathrm{Sym}(\Omega \cup \{\star, \infty\})$ fixes $\star$ and $\infty$, and the cycles $C_1, \ldots, C_r$ of $g$ on $\Omega$ (each fixed point counting as a cycle) satisfy $\omega_i \in C_i$. Then $g\pi$ is the product of two disjoint cycles, of lengths $1 + \sum_{i \text{ odd}} |C_i|$ and $1 + \sum_{i \text{ even}} |C_i|$.*

*Proof.* (a) The $i$-th factor maps $\omega_i \mapsto \star \mapsto \infty \mapsto \omega_i$ and fixes the other $\omega_j$. The point $\omega_i$ is fixed by the factors before the $i$-th; then it goes to $\star$, the $(i+1)$-st factor (if any) sends $\star$ to $\infty$, the $(i+2)$-nd factor (if any) sends $\infty$ to $\omega_{i+2}$, and later factors fix $\omega_{i+2}$. This gives the images of the $\omega_i$. The first factor sends $\star \mapsto \infty$ and the second sends $\infty \mapsto \omega_2$, which is fixed afterwards; and $\infty \mapsto \omega_1$ under the first factor, after which $\omega_1$ is fixed.

(b) Let $z \in C_i$. If $z^g \neq \omega_i$, then $z^g \in C_i \setminus \{\omega_i\}$ is fixed by $\pi$, so $z^{g\pi} = z^g$. If $z^g = \omega_i$, then $z^{g\pi} = \omega_i^\pi$. Moreover $\star^{g\pi} = \omega_2$ and $\infty^{g\pi} = \omega_1$. Hence, starting at $\infty$, the map $g\pi$ goes to $\omega_1$, runs through the whole cycle $C_1$ (following $g$), then jumps to $\omega_1^\pi = \omega_3$, runs through $C_3$, and so on through $C_{r-1}$, after which it returns to $\omega_{r-1}^\pi = \infty$. This is a cycle of length $1 + \sum_{i \text{ odd}} |C_i|$. Likewise the cycle of $\star$ runs through $C_2, C_4, \dots, C_r$. These two cycles exhaust $\Omega \cup \{\star, \infty\}$. □

**Theorem 6.3.** *In Setup 4.1, let $L, U, T, X_0, X_1, Q \in S_n$ be the permutations of Proposition 4.3; they lie in* $\mathrm{Sym}([0, N-1])$. *Let $C, E$ be as in (4) and put*

$$W := \mathrm{Tel}(X_0, E, p)\,\mathrm{Tel}(X_1, E, p).$$

*Then the following* 27 *relators on the letters $L, U, T, X_0, X_1, Q$ form a presentation of the group* $\mathrm{Sym}([0, N-1]) \cong S_N$ *on these six permutations:*

*(R1) the ten relators of Corollary 5.4;*

*(R2)* $X_0^3$, $X_1^3$;

*(R3)* $[X_0, L]$, $[X_0, U]$, $[X_1, L]$, $[X_1, U]$;

*(R4)* $(X_0X_0^T)^2$, $(X_1X_1^T)^2$, $(X_0X_1)^2$, $(X_0X_1^T)^2$;

*(R5)* $(L\,T\,W)^{q+1}$;

*(R6)* $Q^2$, $[Q, L]$, $[Q, U]$, $[Q, T]$, $(QX_0)^2$, $(QX_1)^2$.

*These relators form a uniform set with exponents at most $q+1$.*

*Proof.* Write $o_0 = \iota_0(0) = 0$, $o_1 = \iota_1(0) = q$, and $\Omega = G_0 \cup G_1$. Let $\theta$ be as in Setup 4.1; by Proposition 4.3, $L, U, T$ are $\theta$ of $\alpha_{M_L,0}, \alpha_{M_U,0}, t_{e_1}$, and by Lemma 4.4, $E = \theta(t_{e_2})$.

*Step 1: the relators hold.* (R1) holds by Corollary 5.4, since $\theta$ is a homomorphism. (R2) is clear. (R3): a permutation fixing $\alpha, \beta, \gamma$ commutes with $(\alpha, \beta, \gamma)$, and $\theta(\alpha_{M,0})$ fixes $o_0, o_1, \star, \infty$. (R4): for distinct $\alpha, \beta \in \Omega$ the product $(\alpha, \star, \infty)(\beta, \star, \infty)$ equals $(\alpha, \infty)(\beta, \star)$, an involution; and $X_0^T = (\iota_0(e_1), \star, \infty)$, $X_1^T = (\iota_1(e_1), \star, \infty)$. (R6): $Q$ commutes with $\theta(\mathrm{ASL}(2,p))$, which fixes $\star, \infty$, and $QX_j = (o_j, \star)$ is an involution. (R5): by Lemma 3.1, and because $X_j^{E^b} = (\iota_j(be_2), \star, \infty)$,

$$W = \prod_{b=0}^{p-1} (\iota_0(0,b), \star, \infty) \prod_{b=0}^{p-1} (\iota_1(0,b), \star, \infty).$$

Put $\phi = \alpha_{M_L,0} t_{e_1}$, so $(a,b)^\phi = (a+1, a+b)$ and $LT = \theta(\phi)$. By induction, $(0,b)^{\phi^k} = (k, b + k(k-1)/2)$; since $p$ is odd, $\phi^p$ fixes $(0,b)$, and the first coordinates $0, 1, \dots, p-1$ show that the $\phi$-orbit of $(0,b)$ has length $p$ and meets the line $\{0\} \times \mathbb{F}_p$ only in $(0,b)$. Hence $\phi$ has exactly $p$ cycles, each of length $p$, with representatives $(0,b)$, $b \in \mathbb{F}_p$. Therefore $LT$ fixes $\star, \infty$, and its cycles on $\Omega$ are $2p$ cycles of length $p$, the points $\iota_0(0,b)$, $\iota_1(0,b)$ lying in distinct ones. By Lemma 6.2(b) (with $r = 2p$), $LTW$ is a product of two cycles of length $1 + p \cdot p = q+1$, so (R5) holds.

Let $\Gamma$ be the group defined by the presentation, $\varepsilon\colon \Gamma \to \mathrm{Sym}([0, N-1])$ the homomorphism given by Step 1, $H = \langle L, U, T \rangle \leq \Gamma$ and $\Gamma_A = \langle L, U, T, X_0, X_1 \rangle \leq \Gamma$.

*Step 2: $H \cong \mathrm{ASL}(2,p)$.* By (R1) and Corollary 5.4, $H$ is a quotient of $\mathrm{ASL}(2,p)$, and $\varepsilon(H) = \theta(\mathrm{ASL}(2,p)) \cong \mathrm{ASL}(2,p)$. Hence $\varepsilon|_H$ is injective, and $\vartheta := (\varepsilon|_H)^{-1} \circ \theta\colon \mathrm{ASL}(2,p) \to H$

is an isomorphism with $\vartheta(\alpha_{M_L,0}) = L$, $\vartheta(\alpha_{M_U,0}) = U$, $\vartheta(t_{e_1}) = T$. Through $\vartheta$, the group $H$ acts on $\mathbb{F}_p^2$ (on the right), and $v^h \in \mathbb{F}_p^2$ is defined for $h \in H$. The stabiliser of 0 in $H$ is $\vartheta(\mathrm{SL}(2,p)) = \langle L, U\rangle$, because $\mathrm{SL}(2,p)$ is generated by $M_L$ and $M_U$.

*Step 3: the elements $x_{j,v}$.* For $j \in \{0,1\}$ and $v \in \mathbb{F}_p^2$ put $x_{j,v} := X_j^h$, where $h \in H$ satisfies $0^h = v$. This does not depend on $h$: if $0^h = 0^{h'}$ then $h'h^{-1} \in \langle L, U\rangle$, which centralises $X_j$ by (R3). Clearly $x_{j,v}^g = x_{j,v^g}$ for $g \in H$, $x_{j,0} = X_j$, $x_{0,e_1} = X_0^T$ and $x_{1,e_1} = X_1^T$; and $\varepsilon(x_{j,v}) = (\iota_j(v), \star, \infty)$.

*Step 4: Carmichael's relations.* We claim that $x_{j,v}^3 = 1$ for all $(j,v)$ and $(x_{j,v}x_{j',v'})^2 = 1$ whenever $(j,v) \neq (j',v')$. The first holds by (R2). For the second, recall that $\mathrm{ASL}(2,p)$ is 2-transitive on $\mathbb{F}_p^2$ (translations are transitive and $\mathrm{SL}(2,p)$ is transitive on nonzero vectors). If $j = j'$, choose $g \in H$ with $0^g = v$ and $e_1^g = v'$; then $x_{j,v}x_{j,v'} = (X_jX_j^T)^g$, and (R4) applies. If $j = 0$, $j' = 1$ and $v = v'$, choose $g$ with $0^g = v$; then $x_{0,v}x_{1,v} = (X_0X_1)^g$. If $j = 0$, $j' = 1$ and $v \neq v'$, choose $g$ with $0^g = v$ and $e_1^g = v'$; then $x_{0,v}x_{1,v'} = (X_0X_1^T)^g$. The case $j = 1$, $j' = 0$ follows by the note after Theorem 6.1. By Theorem 6.1, applied with $M = 2q$ to the generators $x_{j,v}$ (in any order), the subgroup $K = \langle x_{j,v} : j \in \{0,1\}, v \in \mathbb{F}_p^2\rangle$ is a homomorphic image of $A_{2q+2} = A_N$.

*Step 5: $\Gamma_A = K$.* The set of the $x_{j,v}$ is invariant under conjugation by $H$, and $X_0, X_1 \in K$; hence $K \trianglelefteq \Gamma_A$ and $\Gamma_A = HK$. By Lemma 3.1, in $\Gamma$ we have $W = \prod_b X_0^{E^b} \prod_b X_1^{E^b} \in K$, as $E \in H$. Put $g = LT \in H$. By Lemma 3.2 and (R5), $1 = (gW)^{q+1} = g^{q+1}\prod_i W^{g^i}$, and every $W^{g^i}$ lies in $K$; hence $g^{q+1} \in K$. In $H \cong \mathrm{ASL}(2,p)$ the element $g = \vartheta(\phi)$ has order $p$ (Step 1), and $q+1 \equiv 1 \pmod p$, so $g \in K$. Now $H \cap K \trianglelefteq H$ contains $g = \vartheta(\alpha_{M_L,e_1})$, whose linear part $M_L$ is not $\pm I$; by Lemma 5.5, $H \cap K = H$. Thus $H \leq K$ and $\Gamma_A = K$.

*Step 6: $\Gamma_A \cong A_N$.* By Steps 4 and 5, $|\Gamma_A| \leq |A_N|$. On the other hand, $\varepsilon(\Gamma_A)$ contains all 3-cycles $(\omega, \star, \infty)$ with $\omega \in \Omega$, which generate $A_N$, and $\varepsilon(\Gamma_A) \leq A_N$: the 3-cycles $X_0, X_1$ are even, and for $\alpha \in \mathrm{ASL}(2,p)$ the restrictions of $\theta(\alpha)$ to $G_0$ and to $G_1$ are conjugate via $\iota_1 \circ \iota_0^{-1}$, hence have the same sign, so $\theta(\alpha)$ is even. Hence $\varepsilon|_{\Gamma_A}$ is an isomorphism onto $A_N$.

*Step 7: $\Gamma \cong S_N$.* By (R6), $Q^2 = 1$, $Q$ commutes with $L, U, T$, and $QX_jQ^{-1} = X_j^{-1}$. Hence $Q$ normalises $\Gamma_A$, so $\Gamma_A \trianglelefteq \Gamma = \Gamma_A\langle Q\rangle$ and $|\Gamma| \leq 2|A_N| = |S_N|$. Since $\varepsilon(Q)$ is odd, $\varepsilon(\Gamma) = S_N$, and $\varepsilon$ is an isomorphism.

Uniformity is clear, since the shapes of the relators do not depend on $p$. □

**Proposition 6.4.** *In the situation of Theorem 6.3 there are PEs $\Xi_p$ and $\Upsilon_p$ over the letters $L, U, T, X_0, X_1, Q$, forming a uniform set (as $p$ varies) with exponents at most $p$, whose values are $(0,1)$ and $(0,1,\dots,N-1)$.*

*Proof.* Let $E$ be as in (4). For $v = (a,b) \in \mathbb{F}_p^2$ let $h_v$ be the PE $T^{\bar a}E^{\bar b}$, where factors with exponent 0 are omitted and an exponent 1 is not written; if $v = 0$, put $x_{j,0} := X_j$, and otherwise $x_{j,v} := X_j^{h_v}$. By Proposition 4.3 and Lemma 4.4, $h_v$ has value $\theta(t_v)$, so $x_{j,v}$ has value $(\iota_j(v), \star, \infty)$. Put $u_{j,v} := x_{j,v}Q$, with value $(\iota_j(v), \infty)$, and for two pairs let

$$[j,v \,|\, j',v'] := u_{j,v}\, u_{j',v'}\, u_{j,v}, \quad \text{with value } (\iota_j(v), \iota_{j'}(v')).$$

Write $t_i = (i, i+1)$. Define

$$R^{(0)} = \mathrm{Tel}\big([0,(0,0)\,|\,0,(0,1)],\, E,\, p-1\big), \qquad J^{(0)} = [0,(0,p-1)\,|\,0,(1,0)],$$
$$R^{(1)} = \mathrm{Tel}\big([1,(0,0)\,|\,1,(0,-1)],\, E^{-1},\, p-1\big), \qquad J^{(1)} = [1,(0,1)\,|\,1,(1,0)],$$
$$\Gamma_j = \mathrm{Tel}\big(R^{(j)}J^{(j)},\, T,\, p-1\big)\,\big(R^{(j)}\big)^{T^{p-1}} \ (j=0,1), \qquad K_1 = [0,(p-1,p-1)\,|\,1,(0,0)],$$
$$K_2 = Q\, x_{1,(p-1,1)},$$

and $\Xi_p := [0,(0,0)\,|\,0,(0,1)]$, $\Upsilon_p := (\Gamma_0K_1\Gamma_1K_2Q)^{-1}$.

Since $\iota_0(0,0) = 0$ and $\iota_0(0,1) = 1$, $\Xi_p$ has value $(0,1)$. The value of $E^b$ maps $\iota_0(0,0) \mapsto \iota_0(0,b) = b$, so by Lemma 3.1 $R^{(0)}$ has value $t_0t_1\cdots t_{p-2}$; and $J^{(0)}$ has value $(p-1,p) = t_{p-1}$.

The value of $T^a$ maps $\iota_0(a',b) \mapsto \iota_0(a'+a,b)$, i.e. it shifts rows of $G_0$; so for $0 \le a \le p-2$ the value of $(R^{(0)}J^{(0)})^{T^a}$ is $t_{ap}t_{ap+1}\cdots t_{ap+p-1}$, and the value of $(R^{(0)})^{T^{p-1}}$ is $t_{(p-1)p}\cdots t_{q-2}$. By Lemma 3.1, $\Gamma_0$ has value $t_0t_1\cdots t_{q-2}$. Next $K_1$ has value $(\iota_0(p-1,p-1),\iota_1(0,0)) = (q-1,q) = t_{q-1}$. In $G_1$ we have $\iota_1(0,-b) = q+b$ for $0 \le b \le p-1$, and the value of $E^{-b}$ maps $\iota_1(0,0) \mapsto \iota_1(0,-b)$ and $\iota_1(0,-1) \mapsto \iota_1(0,-1-b)$; hence $R^{(1)}$ has value $t_qt_{q+1}\cdots t_{q+p-2}$. Further $J^{(1)}$ has value $(\iota_1(0,1),\iota_1(1,0)) = (q+p-1,q+p) = t_{q+p-1}$, and $T$ shifts the rows of $G_1$ as well; so $\Gamma_1$ has value $t_qt_{q+1}\cdots t_{2q-2}$. Finally $K_2$ has value $(\star,\infty)(\iota_1(p-1,1),\star,\infty) = (\iota_1(p-1,1),\star) = (2q-1,2q) = t_{2q-1}$, and $Q$ has value $(2q,2q+1) = t_{2q}$. Altogether $\Gamma_0K_1\Gamma_1K_2Q$ has value $t_0t_1\cdots t_{N-2} = (0,N-1,N-2,\ldots,1)$ by Lemma 3.3, so $\Upsilon_p$ has value $(0,1,\ldots,N-1)$. The shapes of these PEs do not depend on $p$ (apart from the omission of factors with exponent 0 or 1, which happens only in finitely many ways), so they form a uniform set; all exponents are at most $p$. □

# 7 Gluing two copies

**Lemma 7.1.** *For every integer $n \ge 245$ there is a prime $p \ge 11$ such that $N = 2p^2+2$ satisfies $\frac{n+3}{2} \le N \le n-1$.*

*Proof.* The condition says $n \in I_p := [2p^2+3,\, 4p^2+1]$. We have $\min I_{11} = 245$, so it suffices to show that $I_p \cup I_{p'}$ is an interval whenever $p < p'$ are consecutive primes with $p \ge 11$, i.e. that $2p'^2+3 \le 4p^2+2$, i.e. $p'^2 \le 2p^2-1$. For $p \in \{11,13,17,19,23\}$ (with $p' = 13,17,19,23,29$) this is checked directly: $169 \le 241$, $289 \le 337$, $361 \le 577$, $529 \le 721$, $841 \le 1057$. For $p \ge 29$, Nagura's theorem [10] (for every real $\xi \ge 25$ there is a prime in $(\xi, \frac{6}{5}\xi)$) gives $p' < \frac{6}{5}p$, so $p'^2 < 1.44\,p^2 \le 2p^2-1$. □

**Setup 7.2.** Let $n \ge 245$, choose $p$ as in Lemma 7.1, and put $q = p^2$, $N = 2q+2$, $s = n-N$ and $k = 2N-n$. Then $1 \le s \le N-3$ and $k \ge 3$. Let $L,\ldots,Q \in S_n$ and the PEs $W_L,\ldots,W_Q$ over $\{x,y\}$ be as in Definition 4.2 and Proposition 4.3. Consider the alphabet

$$\mathcal{A} = \{x,y\} \sqcup \{L^{(1)},U^{(1)},T^{(1)},X_0^{(1)},X_1^{(1)},Q^{(1)}\} \sqcup \{L^{(2)},U^{(2)},T^{(2)},X_0^{(2)},X_1^{(2)},Q^{(2)}\}$$

and the map $\varphi\colon \mathcal{A} \to S_n$ with $\varphi(x) = x$, $\varphi(y) = y$, $\varphi(G^{(1)}) = G$ and $\varphi(G^{(2)}) = G^{y^s}$ for $G \in \{L,U,T,X_0,X_1,Q\}$. Put $I_1 = [0,N-1]$, $I_2 = [s,n-1]$, $a_1 = 0$, $a_2 = s$. By Lemma 3.4(e), $\varphi(G^{(2)})$ is $\varphi(G^{(1)})$ shifted by $s$, so by Theorem 6.3 the six permutations $\varphi(G^{(\kappa)})$ generate $\mathrm{Sym}(I_\kappa)$ for $\kappa = 1,2$. For $\kappa = 1,2$ let $\Pi^{(\kappa)}$ be the 27 relators of Theorem 6.3 written in the letters $G^{(\kappa)}$, and let $\Xi^{(\kappa)},\Upsilon^{(\kappa)}$ be the PEs of Proposition 6.4 written in these letters; their values are $(a_\kappa, a_\kappa+1)$ and $(a_\kappa, a_\kappa+1,\ldots,a_\kappa+N-1)$. For $3 \le m \le N$ define

$$\rho_m^{(\kappa)} = \Big((\Xi^{(\kappa)}(\Upsilon^{(\kappa)})^{-1})^{m-1}(\Upsilon^{(\kappa)})^{m-1}\Big)^{-1}.$$

Applying Lemma 3.4 inside $\mathrm{Sym}(I_\kappa) \cong S_N$ (with $\Xi^{(\kappa)},\Upsilon^{(\kappa)}$ in the roles of $x,y$), the value of $\rho_m^{(\kappa)}$ is $(a_\kappa, a_\kappa+1,\ldots,a_\kappa+m-1)$, and if a PE $F$ has a value supported in $[a_\kappa, a_\kappa+N-1-r]$, then $F^{(\Upsilon^{(\kappa)})^r}$ has as value the value of $F$ shifted by $r$.

**Theorem 7.3.** *In Setup 7.2, consider the following relators over $\mathcal{A}$:*

*(G1)* $\Pi^{(1)}$ *and* $\Pi^{(2)}$ *(54 relators);*

*(G2)* $\varsigma^{(1)}(\varsigma^{(2)})^{-1}$ *and* $\gamma^{(1)}(\gamma^{(2)})^{-1}$, *where* $\varsigma^{(1)} = (\Xi^{(1)})^{(\Upsilon^{(1)})^s}$, $\gamma^{(1)} = (\rho_k^{(1)})^{(\Upsilon^{(1)})^s}$, $\varsigma^{(2)} = \Xi^{(2)}$ *and* $\gamma^{(2)} = \rho_k^{(2)}$;

*(G3)* $[\ell, r]$ *for all* $\ell \in \Lambda$ *and* $r \in \mathcal{R}$*, where, if* $s \geq 2$*,*

$$\Lambda = \{\Xi^{(1)}, \rho_{s+1}^{(1)}\}, \qquad \mathcal{R} = \big\{(\Xi^{(2)})^{(\Upsilon^{(2)})^{N-1-s}},\ (\rho_{s+1}^{(2)})^{(\Upsilon^{(2)})^{N-1-s}}\big\},$$

*and if* $s = 1$*,* $\Lambda = \{\Xi^{(1)}\}$ *and* $\mathcal{R} = \{(\Xi^{(2)})^{(\Upsilon^{(2)})^{N-2}}\}$*;*

*(G4)* $x(\Xi^{(1)})^{-1}$ *and* $y\,(r^*\,\Upsilon^{(1)})^{-1}$*, where* $r^*$ *is the last element of* $\mathcal{R}$ *as listed.*

*This is a presentation of* $S_n$ *via* $\varphi$*, with* 62 *relators if* $s \geq 2$ *and* 59 *if* $s = 1$*.*

*Proof. Step 1: values.* The relators (G1) hold by Theorem 6.3 (for $\kappa = 2$ transported by conjugation with $y^s$). For (G2): $(0,1)$ is supported in $[0,1] \subseteq [0, N-1-s]$ and $(0,\dots,k-1)$ in $[0,k-1] = [0,N-1-s]$, so $\varsigma^{(1)}$ and $\gamma^{(1)}$ have values $(s,s+1)$ and $(s,s+1,\dots,N-1)$, which are also the values of $\varsigma^{(2)}$ and $\gamma^{(2)}$. For (G3): the elements of $\Lambda$ have values $(0,1)$ and $(0,1,\dots,s)$, generating $\mathrm{Sym}([0,s])$; for $\mathcal{R}$, the values $(s,s+1)$ and $(s,\dots,2s)$ are supported in $[a_2, a_2+N-1-(N-1-s)] = [s,2s]$, so after the shift by $N-1-s$ they become $(N-1,N)$ and $(N-1,N,\dots,n-1)$, which generate $\mathrm{Sym}([N-1,n-1])$ (if $s=1$, $\mathrm{Sym}([N-1,n-1]) = \langle(N-1,N)\rangle$). Since $s < N-1$, the sets $[0,s]$ and $[N-1,n-1]$ are disjoint, so the commutators in (G3) are trivial. For (G4): $\varphi(r^*) = (N-1,N,\dots,n-1)$, and $(N-1,N,\dots,n-1)(0,1,\dots,N-1)$ maps $i \mapsto i+1$ for $i \leq n-2$ and $n-1 \mapsto N-1 \mapsto 0$; so it equals $y$.

Let $\Delta$ be the group defined by the presentation and $\Delta_\kappa = \langle G^{(\kappa)}\rangle \leq \Delta$. By (G4), $\Delta = \langle \Delta_1, \Delta_2\rangle$.

*Step 2.* By (G1) and Theorem 6.3, $\Delta_\kappa$ is a homomorphic image of $S_N$, and $\varphi(\Delta_\kappa) = \mathrm{Sym}(I_\kappa)$ has order $N!$. Hence $\varphi|_{\Delta_\kappa}\colon \Delta_\kappa \to \mathrm{Sym}(I_\kappa)$ is an isomorphism.

*Step 3: adjacent transpositions.* For $0 \leq i \leq N-2$ let $s_i \in \Delta_1$ and for $s \leq i \leq n-2$ let $s_i' \in \Delta_2$ be the preimages of $(i,i+1)$ under these isomorphisms. We claim $s_i = s_i'$ for $s \leq i \leq N-2$. Indeed, $O = [s,N-1]$ has $k \geq 3$ points and $\mathrm{Sym}(O)$ is generated by $(s,s+1)$ and $(s,\dots,N-1)$, so $(i,i+1) = w_i\big((s,s+1),(s,\dots,N-1)\big)$ for some word $w_i$. Since $\varphi|_{\Delta_\kappa}$ is an isomorphism, $s_i = w_i(\varsigma^{(1)},\gamma^{(1)})$ and $s_i' = w_i(\varsigma^{(2)},\gamma^{(2)})$, and these are equal by (G2). Define $\tau_i = s_i$ for $0 \leq i \leq N-2$ and $\tau_i = s_i'$ for $s \leq i \leq n-2$.

*Step 4: Coxeter relations.* We show that $\tau_0,\dots,\tau_{n-2}$ satisfy $\tau_i^2 = 1$, $(\tau_i\tau_{i+1})^3 = 1$ and $(\tau_i\tau_j)^2 = 1$ for $j \geq i+2$. The first two hold inside $\Delta_1$ or $\Delta_2$, because for each $i$ either $i+1 \leq N-2$ or $i \geq N-2 \geq s$ (note $N-2 \geq s$ as $2N \geq n+3$). For $(\tau_i\tau_j)^2$ with $i+2 \leq j$: if $j \leq N-2$ or $i \geq s$ it holds in $\Delta_1$ or $\Delta_2$. Otherwise $i \leq s-1$ and $j \geq N-1$, so $(i,i+1) \in \mathrm{Sym}([0,s])$ and $(j,j+1) \in \mathrm{Sym}([N-1,n-1])$; hence $\tau_i \in \langle\Lambda\rangle$ and $\tau_j \in \langle\mathcal{R}\rangle$ (as $\varphi|_{\Delta_\kappa}$ are isomorphisms), and these subgroups commute by (G3); so $(\tau_i\tau_j)^2 = \tau_i^2\tau_j^2 = 1$.

*Step 5.* By Moore's theorem [9] (see [4, §6.2]), $S_n$ is presented by generators $\tau_0,\dots,\tau_{n-2}$ and the relations of Step 4. Hence $\langle\tau_0,\dots,\tau_{n-2}\rangle$ is a homomorphic image of $S_n$. It contains $\Delta_1 = \langle s_0,\dots,s_{N-2}\rangle$ and $\Delta_2 = \langle s_s',\dots,s_{n-2}'\rangle$ (adjacent transpositions generate $\mathrm{Sym}(I_\kappa)$), so it equals $\Delta$. Thus $|\Delta| \leq n!$, and since $\varphi(\Delta) \ni x, y$, $\varphi$ induces an isomorphism $\Delta \to S_n$. □

*Proof of Theorem 1.1.* For $n < 245$ there are only finitely many groups $S_n$, each of which has some finite presentation on $x,y$; these determine admissible values of $r_0$ and $c_0$ for $n < 245$. Let $n \geq 245$ and apply Lemma 2.5 to the presentation of Theorem 7.3, with $\mathcal{B} = \mathcal{A} \setminus \{x,y\}$, $W_{G^{(1)}} = W_G$ and $W_{G^{(2)}} = (W_G)^{y^s}$, where $W_G$ is the PE of Definition 4.2 for $G$. This gives a presentation of $S_n$ on $x,y$ with at most 62 relators. The relators of Theorem 7.3 form a uniform set: they are built by a fixed procedure from the uniform sets of Theorem 6.3 and Proposition 6.4, and their shapes take only finitely many forms (depending on whether $s=1$ and on which exponents equal 0 or 1 and are therefore omitted or not written). The PEs $W_{G^{(\kappa)}}$ form a uniform set by Proposition 4.3 and Lemma 3.4. By (2), the substituted relators form a uniform set; all exponents occurring are at most $n$ in absolute value. By Lemma 2.2, the length of the resulting presentation is at most $c(1+\log_2 n) \leq c_0 \log n$ for an absolute constant $c_0$. □

## 8 Remarks and computational checks

**Remark 8.1.** (a) The constants are large. For the presentations constructed above we computed the length after elimination: about $4.1 \times 10^6$ for $n \approx 10^3$, $8.6 \times 10^6$ for $n \approx 10^9$ and $2.0 \times 10^7$ for $n \approx 10^{24}$; the ratio $\ell / \log_2 n$ decreases slowly to about $2.5 \times 10^5$. The nesting depth of the relators is 17. No attempt was made to reduce these numbers.

(b) For the alternating groups, one may apply the Reidemeister–Schreier argument of [2, §4] to the presentation of Theorem 1.1. When applied to power expressions, the rewriting can double the length at each level of nesting, so, as the nesting depth is bounded, the length is multiplied by at most a constant factor. We have not written out the details, and we do not claim this as part of the results of this note.

All formulas in this note were checked by a computer program, `verify_paper.py`, which accompanies it and implements the definitions literally (same names and conventions). It checks:

(1) Lemma 3.4 for several $n$;

(2) all value statements of Proposition 4.3 and Lemma 4.4 for $p \in \{5, 7, 11, 13, 17\}$ and two values of $n$ each;

(3) the relators of Propositions 5.2, 5.3 and Corollary 5.4 in $\mathrm{ASL}(2, p)$ for all primes $5 \le p < 200$; and, by coset enumeration, the orders $|\mathrm{SL}(2, p)|$ and the index $p^2$ for $p \le 23$, the order $|\mathrm{ASL}(2, p)|$ of the presentation of Corollary 5.4 for $p = 5, 7$, and the index $p^2$ of $\langle L, U \rangle$ in it for $p = 11, 13$;

(4) Carmichael's presentation for $M \le 6$ by coset enumeration, and Lemma 6.2(a) in 200 random cases;

(5) all 27 relators of Theorem 6.3, the cycle type of $LTW$, and Proposition 6.4, for all primes $5 \le p \le 23$;

(6) Lemma 7.1 for all $n \le 2 \cdot 10^5$ and the inequality $p'^2 \le 2p^2 - 1$ for consecutive primes below $2 \cdot 10^4$;

(7) all relators of Theorem 7.3 in $S_n$ for 25 values of $n$ between 245 and $6 \cdot 10^4$ (including both cases $s = 1$ and $s \ge 2$), and the lengths quoted above.

These computations are not part of the proofs; they serve to guard against errors in the explicit formulas.

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